

Local AI-Based Adaptive Traffic Signal Control with Emergency Vehicle Priority

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Abstract—Urban traffic congestion is a major challenge caused by increasing vehicle density and changing traffic conditions. Conventional fixed-time traffic signals cannot always respond effectively to real-time traffic demand. This paper proposes a Local AI-Based Adaptive Traffic Signal Control system with Emergency Vehicle Priority. The system processes traffic information near the intersection, estimates traffic density, and dynamically selects signal phases. An emergency-priority mechanism is included to provide temporary priority when an emergency vehicle is detected. Local processing reduces dependence on continuous cloud communication. The paper presents the proposed architecture, methodology, decision algorithm, mathematical model, implementation concept, limitations, and future scope.

Index Terms—Local AI, adaptive traffic signals, traffic management, emergency vehicle priority, edge computing

I. INTRODUCTION

Urbanization and increasing vehicle ownership have resulted in significant traffic congestion. Traffic intersections are particularly important because inefficient signal timing can increase vehicle waiting time and reduce road capacity.

Traditional traffic signals generally use fixed timing schedules. Such systems cannot always respond to sudden changes in traffic density. For example, one road may experience heavy traffic while another remains relatively empty.

Artificial intelligence provides an opportunity to develop adaptive traffic signal systems. Reinforcement learning has been investigated for traffic signal control [1], while machine vision has been used to obtain traffic information for adaptive signal operation [2]. IntelliLight also demonstrated the application of reinforcement learning to intelligent traffic-light control [3].

This work proposes a local AI architecture that analyzes traffic conditions near an intersection and dynamically selects signal phases. Emergency vehicle priority is also incorporated to support emergency movement.

II. LITERATURE REVIEW

Rasheed et al. reviewed deep reinforcement learning techniques for traffic signal control and discussed their application to adaptive traffic management [1].

Gorodokin et al. investigated adaptive traffic-light control using machine vision, demonstrating the usefulness of visual traffic information for signal control [2].

Wei et al. proposed IntelliLight, a reinforcement-learning approach for intelligent traffic-light control based on traffic-state information [3].

These studies establish the potential of AI-based traffic control. However, combining local processing, adaptive signal control, and emergency-vehicle priority provides an additional direction for investigation.

III. PROBLEM STATEMENT

Fixed-time traffic signals have limited ability to respond to changing traffic density. Centralized intelligent systems may also depend on continuous communication with remote servers. Emergency vehicles can experience delays when normal signal cycles are maintained.

Therefore, this work addresses the problem of designing a local AI-based traffic signal system that can adapt to traffic conditions while providing temporary priority to detected emergency vehicles.

IV. RESEARCH GAP

Existing research has demonstrated AI-based traffic signal control using reinforcement learning and machine vision [1]–[3]. However, there is an opportunity to combine:

- local traffic analysis,
- adaptive signal timing,
- emergency-vehicle priority, and
- reduced dependence on cloud processing.

The proposed work focuses on this combination at the intersection level.

V. OBJECTIVES

The objectives of the proposed system are:

- 1) Estimate traffic density using local traffic information.
- 2) Dynamically select appropriate signal phases.
- 3) Detect emergency-vehicle situations.
- 4) Provide temporary emergency priority.
- 5) Reduce dependence on continuous cloud communication.

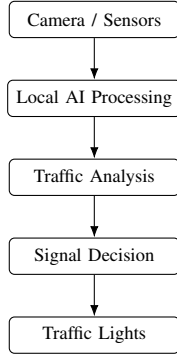


Fig. 1. Local AI-based traffic signal architecture.

VI. PROPOSED SYSTEM

The proposed system uses traffic cameras or sensors to obtain information about vehicles approaching an intersection. A local AI processing unit analyzes the information and estimates traffic density.

Under normal conditions, the controller selects a signal phase based on traffic demand. If an emergency vehicle is detected, the emergency-priority module temporarily modifies the signal sequence.

The main modules are:

- Traffic sensing
- Local AI processing
- Traffic-density estimation
- Emergency-vehicle detection
- Signal decision
- Traffic-light control

VII. SYSTEM ARCHITECTURE

The proposed architecture is shown in Fig. 1.

Traffic information is captured by cameras or sensors. The local processing unit analyzes the input and estimates traffic conditions. The decision module then selects an appropriate signal phase.

VIII. SYSTEM FLOW

The operating procedure of the proposed system is shown in Fig. 2.

The system continuously collects traffic information, processes it locally, checks for an emergency vehicle, and selects either adaptive signal timing or emergency priority.

IX. METHODOLOGY

The methodology consists of five stages.

A. Traffic Data Collection

Cameras or sensors collect information about vehicles approaching the intersection.

B. Local Processing

The collected information is processed near the intersection instead of continuously transmitting raw data to a remote server.

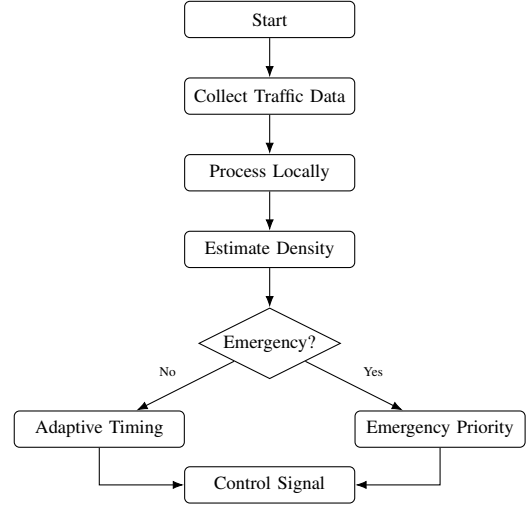


Fig. 2. Compact flowchart of the proposed system.

C. Traffic Density Estimation

Traffic density for road i can be represented as:

$$D_i = \frac{N_i}{L_i} \quad (1)$$

where N_i is the number of detected vehicles and L_i is the effective observation length.

D. Adaptive Signal Decision

The controller assigns priority according to traffic demand while respecting minimum and maximum signal timing constraints.

E. Emergency Priority

An emergency indicator is defined as:

$$E_i = \begin{cases} 1, & \text{emergency vehicle detected,} \\ 0, & \text{otherwise.} \end{cases} \quad (2)$$

When $E_i = 1$, the controller activates the emergency-priority procedure.

X. ALGORITHM

The basic decision procedure is given in Algorithm 1.

XI. MATHEMATICAL MODEL

The traffic demand score for road i can be represented as:

$$T_i = w_d D_i + w_q Q_i \quad (3)$$

where D_i represents traffic density, Q_i represents queue length, and w_d and w_q are weighting factors.

The selected road can be represented by:

$$i^* = \arg \max_i (T_i). \quad (4)$$

During an emergency, the emergency-priority decision overrides the normal traffic-demand decision subject to safety constraints.

Algorithm 1 Local AI Traffic Signal Control

```
1: Capture traffic data
2: Process data locally
3: Estimate traffic density
4: Check for emergency vehicle
5: if emergency vehicle detected then
6:   Activate emergency priority
7: else
8:   Select phase using traffic density
9: end if
10: Update traffic signal
11: Repeat
```

TABLE I
CONCEPTUAL COMPARISON

Feature	Fixed-Time	Local AI
Traffic adaptation	Limited	Dynamic
Traffic analysis	Predefined	Real-time
Signal timing	Fixed	Adaptive
Emergency priority	Separate	Integrated
Cloud dependence	Varies	Reduced

XII. IMPLEMENTATION CONCEPT

A prototype can use a camera, an edge-computing device, and a microcontroller-based traffic-light interface. A lightweight object detection model can estimate vehicle count and identify emergency vehicles.

A simplified control program is shown below.

Listing 1. Simplified traffic-control logic.

```
def controller(density, emergency):
    if emergency:
        return "EMERGENCY PRIORITY"

    if density == "HIGH":
        return "LONG GREEN"
    elif density == "MEDIUM":
        return "NORMAL GREEN"
    else:
        return "SHORT GREEN"

print(controller("HIGH", False))
```

The example represents the decision logic; a real implementation would obtain the inputs from sensors or computer-vision models.

XIII. CONCEPTUAL COMPARISON

The comparison describes the intended functional characteristics of the proposed architecture and does not represent measured experimental performance.

XIV. RESULTS AND DISCUSSION

The proposed architecture is designed to adapt signal decisions according to traffic conditions. Under normal conditions, roads with higher traffic demand can receive appropriate priority.

When an emergency vehicle is detected, the emergency-priority mechanism can temporarily modify the normal signal sequence. Local processing can also reduce the need to continuously transmit raw traffic information to a remote server.

Actual improvements in waiting time, traffic throughput, detection accuracy, and emergency response time must be established through experimental testing. Therefore, no numerical performance claims are made in this conceptual study.

XV. EVALUATION METRICS

The proposed system can be evaluated using:

- Average vehicle waiting time.
- Average queue length.
- Traffic throughput.
- Signal decision latency.
- Emergency-vehicle waiting time.
- Detection accuracy.
- Processing-resource usage.

These measurements can be compared with a conventional fixed-time controller.

XVI. LIMITATIONS

The proposed system is currently a conceptual architecture and does not report experimentally measured performance.

Camera-based detection may be affected by lighting, weather, occlusion, and camera position. Emergency-vehicle detection may also produce false positive or false negative results.

Local AI processing requires suitable hardware. Complex models may need optimization before deployment on resource-constrained edge devices.

Traffic-signal control is safety-critical; therefore, a real deployment requires extensive testing, fail-safe mechanisms, and appropriate regulatory approval.

XVII. FUTURE SCOPE

Future work can include implementation using real traffic data and edge hardware. Lightweight computer-vision models can be evaluated for vehicle and emergency-vehicle detection.

Reinforcement learning can also be investigated for adaptive signal optimization. Multiple intersections can exchange limited information to coordinate emergency routes.

Pedestrian detection, cyclist detection, weather awareness, and secure communication can also be incorporated into future versions.

XVIII. CONCLUSION

This paper proposed a Local AI-Based Adaptive Traffic Signal Control system with Emergency Vehicle Priority. The architecture combines local traffic analysis, adaptive signal timing, and emergency-vehicle prioritization.

The proposed approach can respond to changing traffic conditions while reducing dependence on continuous cloud processing. The emergency-priority mechanism provides a framework for temporarily modifying signal operation when an emergency vehicle is detected.

The work provides a foundation for future implementation and experimental evaluation using real traffic data, edge-computing hardware, and suitable AI models.

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